

ESP Zigbee Gateway

This tutorial will guide you on how to run the ESP Zigbee Gateway example program using the Unit Gateway H2 paired with the CoreS3 controller. The ESP Zigbee Gateway is a gateway device based on the ESP32 series Wi-Fi SoC and the ESP32-H2 802.15.4 SoC. It connects Zigbee networks with Wi-Fi networks, enabling interoperability between smart home devices.

1. Preparation

- 1. **Environment Setup:** Refer to the [ESP-IDF - ESP32S3 Getting Started Tutorial](#) to complete the basic compilation environment setup.

ESP-IDF version

It is recommended to use ESP-IDF version **v5.3.1** for compiling this example.

```
git clone -b v5.3.1 --recursive https://github.com/espressif/esp-idf.git
cd esp-idf
./install.sh
. ./export.sh
```

- 2. Use the `git clone --recursive` command to recursively clone the ESP-Zigbee-SDK repository:

```
git clone --recursive https://github.com/espressif/esp-zigbee-sdk.git
cd esp-zigbee-sdk
```

- 3. **Hardware Products Used:**

- [Unit Gateway H2](#)
- [CoreS3](#)



- **4. The `idf.py` commands used in the subsequent tutorials depend on ESP-IDF. Before running the commands, you need to execute `./export.sh` in ESP-IDF to activate the relevant environment variables. For detailed instructions, please refer to the [ESP-IDF - ESP32S3 Getting Started Tutorial](#).

2. Compile RCP Firmware

- 1. Before compiling the Gateway firmware, you need to generate the RCP firmware first. Refer to the following commands to enter the corresponding RCP firmware directory and set the compilation target to `esp32h2`:

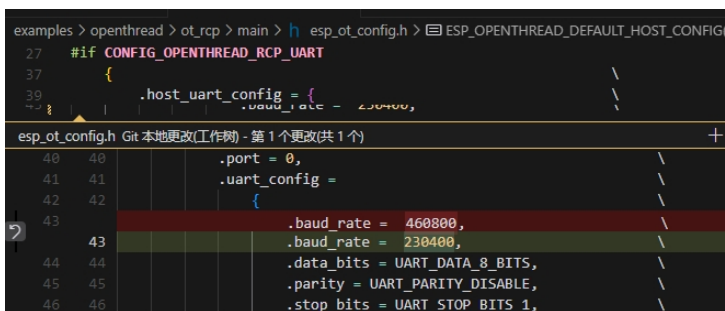
```
cd $IDF_PATH/examples/openthread/ot_rcp
idf.py set-target esp32h2
idf.py menuconfig
```

- 2. Use `idf.py menuconfig` to enter the configuration page. In `menuconfig`, configure: Component config -> OpenThread RCP Example - Enable `OPENTHREAD_NCP_VENDOR_HOOK`.

Baud Rate Modification

Due to factors such as the pin drive capability of the chip, the serial port may not work properly when using a long connecting cable with the Unit. If this happens, you need to reduce the speed appropriately. Here, the default baud rate of 460800 is reduced to 230400.

- 3. Modify the baud rate by opening `main/esp_ot_config.h` and changing line 43 `.baud_rate = 460800`; to `.baud_rate = 230400`;



```
examples > openthread > ot_rcp > main > h esp_ot_config.h > ESP_OPENTHREAD_DEFAULT_HOST_CONFIG
27 #if CONFIG_OPENTHREAD_RCP_UART
37 {
39     .host_uart_config = {
40         .baud_rate = 460800,
41     },
42     .uart_config = {
43         .baud_rate = 460800,
44         .baud_rate = 230400,
45         .data_bits = UART_DATA_8_BITS,
46         .parity = UART_PARITY_DISABLE,
47         .stop_bits = UART_STOP_BITS_1,
48     },
49 },
50 #endif
```

- 4. After completing the configuration, execute the following command to compile the RCP firmware:

```
idf.py build
```

- 5. Open the Unit Gateway H2 case, press the boot button, and then connect the USB power supply to put it into download mode. Execute the following command to flash the firmware:



```
idf.py flash
```

3. Compile Gateway Firmware

- 1. Enter the `esp_zigbee_gateway` example program path and set the compilation target:

```
cd esp-zigbee-sdk/examples/esp_zigbee_gateway
idf.py set-target esp32s3
idf.py menuconfig
```

- 2. Configure the correct communication pins in `menuconfig`. This pin configuration is for the CoreS3 controller. If you are using a different controller, modify it according to the actual situation:

- Board Configuration
 - Pin to RCP reset: -1
 - Pin to RCP boot: -1
 - Pin to RCP TX: 18
 - Pin to RCP RX: 17

```
(Top) → ESP Zigbee gateway rcp update → Board Configuration
Espressif IoT Development Framework Configuration
(-1) Pin to RCP reset
(-1) Pin to RCP boot
(18) Pin to RCP TX
(17) Pin to RCP RX
```

3. Gateway WiFi Connection Configuration:

- Example Connection Configuration
 - WiFi SSID
 - WiFi Password

```
(Top) → Example Connection Configuration
[*] connect using WiFi interface
[ ]   Get ssid and password from stdin
[*]   Provide wifi connect commands
xxxxxx) WiFi SSID
xxxxxx) WiFi Password
(6)   Maximum retry
      WiFi Scan Method (All Channel) --->
      WiFi Scan threshold --->
      WiFi Connect AP Sort Method (Signal) --->
[ ] connect using Ethernet interface
[ ] connect using Point to Point interface
[*] Obtain IPv6 address
      Preferred IPv6 Type (Local Link Address) --->
```

Add the following functions and header files, and call and initialize them at the beginning of app_main to enable the Grove power supply output capability of CoreS3.

```
#include "driver/i2c.h"

void fix_aw9523_p0_pull_up(void)
{
    /* AW9523 P0 is in push-pull mode */
    const i2c_config_t i2c_conf = {
        .mode = I2C_MODE_MASTER,
        .sda_io_num = GPIO_NUM_12,
        .sda_pullup_en = GPIO_PULLUP_DISABLE,
        .scl_io_num = GPIO_NUM_11,
        .scl_pullup_en = GPIO_PULLUP_DISABLE,
        .master.clk_speed = 400000
    };
    i2c_param_config(I2C_NUM_1, &i2c_conf);
    i2c_driver_install(I2C_NUM_1, i2c_conf.mode, 0, 0, 0);

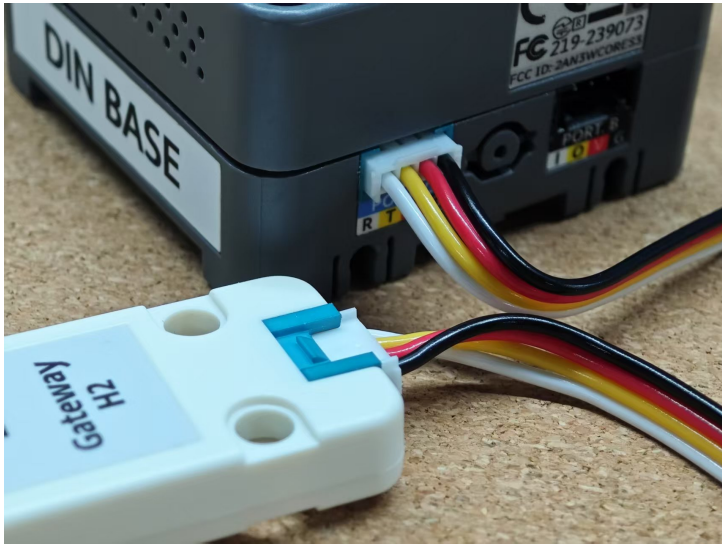
    uint8_t data[2];
    data[0] = 0x11;
    data[1] = 0x10;
    i2c_master_write_to_device(I2C_NUM_1, 0x58, data, sizeof(data), 1000 / portTICK_PERIOD_MS);
    i2c_driver_delete(I2C_NUM_1);
}
```

4. Compile and Flash

```
idf.py build
idf.py erase_flash
idf.py flash
```

5. Start Running

- Connect the CoreS3 with the Unit Gateway H2.
- Connect the CoreS3 to the computer.
- Use `idf.py monitor` or other serial debugging tools to view the running logs at 115200bps.



```
I (3551) wifi:set rx beacon pti, rx_bcn_pti: 0, bcn_timeout: 25000, mt_pti: 0, mt_time: 10000
I (3561) wifi:AP's beacon interval = 163840 us, DTIM period = 1
I (3561) wifi:<ba-add>idx:0 (ifx:0, c4:5e:5c:9b:63:f0), tid:6, ssn:2, winSize:64
I (3611) wifi:<ba-add>idx:1 (ifx:0, c4:5e:5c:9b:63:f0), tid:0, ssn:0, winSize:64
I (4651) esp_netif_handlers: example_netif_sta ip: 192.168.52.21, mask: 255.255.255.0, gw: 192.168.52.1
I (4651) example_connect: Got IPv4 event: Interface "example_netif_sta" address: 192.168.52.21
I (5461) example_connect: Got IPv6 event: Interface "example_netif_sta" address: fe80:0000:0000:0000:4a27:e2ff:fe78:fd74, type: ESP_IP6_ADDR_IS_LINK_LOCAL
I (5461) example_common: Connected to example_netif_sta
I (5461) example_common: - IPv4 address: 192.168.52.21,
I (5471) example_common: - IPv6 address: fe80:0000:0000:0000:4a27:e2ff:fe78:fd74, type: ESP_IP6_ADDR_IS_LINK_LOCAL
I (5481) wifi:Set ps type: 0, coexist: 0

I (5481) ESP_ZB_GATEWAY: Initialize Zigbee stack
W (5511) ESP_ZB_GATEWAY: Network(0xb3f7) closed, devices joining not allowed.
I (5511) ESP_ZB_GATEWAY: Device started up in non factory-reset mode
I (5511) ESP_ZB_GATEWAY: Device rebooted
I (6121) ESP_ZB_GATEWAY: Network(0xb3f7) is open for 180 seconds
```

Normal running log content:

- RCP firmware version check
- Wi-Fi connection successful
- Zigbee network creation successful
- Network open for device joining